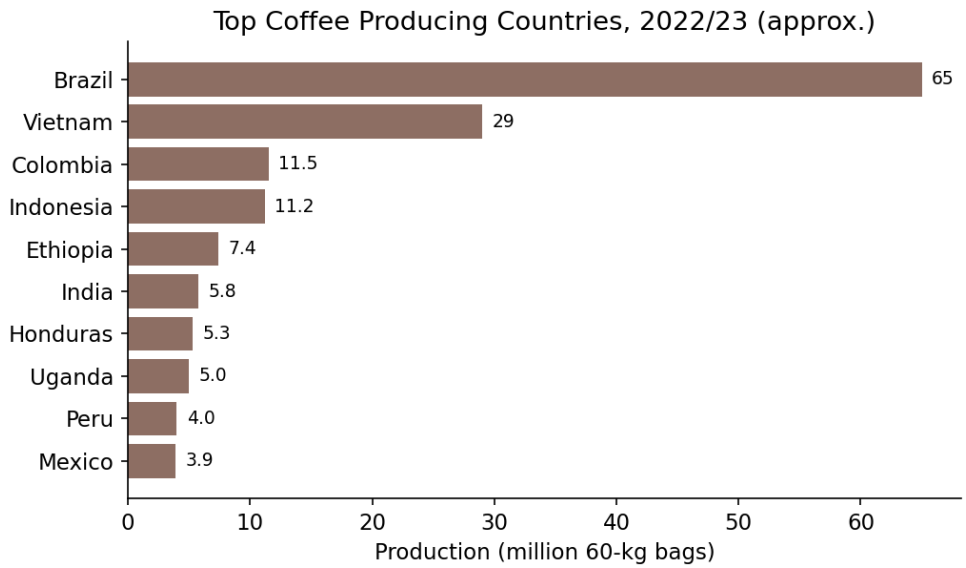
THE GLOBAL COFFEE INDUSTRY

A Reference Guide



From Bean to Cup: Botany, Processing, Roasting, Brewing, Trade, Sustainability and Culture

Prepared as a test document for multimodal retrieval-augmented generation (RAG) systems. Numerical values are rounded approximations for illustration.

Table of Contents

1. Introduction to Coffee 3

2. Botany and Species 5

3. Growing Regions and Production 7

4. From Cherry to Green Coffee: Harvesting and Processing 9

5. Roasting 11

6. Brewing and Preparation 13

7. The Chemistry of Coffee and Health 15

8. Global Trade, Prices and Economics 17

9. Sustainability and Climate Change 19

10. Coffee Culture and Cafés 21

11. Market Trends and the Future of Coffee 23

12. Quality Evaluation and Cupping 25

13. Coffee Traditions Around the World 28

14. Appendix: Worked Examples and Frequently Asked Questions 31

1. Introduction to Coffee

Coffee is one of the most widely traded agricultural commodities and one of the most popular beverages in the world. It is produced from the roasted seeds of plants in the genus Coffea, which grow in tropical regions between the Tropic of Cancer and the Tropic of Capricorn, an area often called the Coffee Belt. More than 2.2 billion cups are consumed worldwide every day, and the industry supports the livelihoods of an estimated 125 million people, most of them smallholder farmers.

According to the legend, coffee was discovered in Ethiopia by a goat herder named Kaldi, who noticed that his goats became energetic after eating the red berries of a certain bush. Historical evidence shows that coffee drinking began in Yemen in the 15th century, where Sufi monasteries used the drink to stay awake during night prayers. From the port of Mocha, coffee spread to Cairo, Mecca, Constantinople and then to Europe in the 17th century.

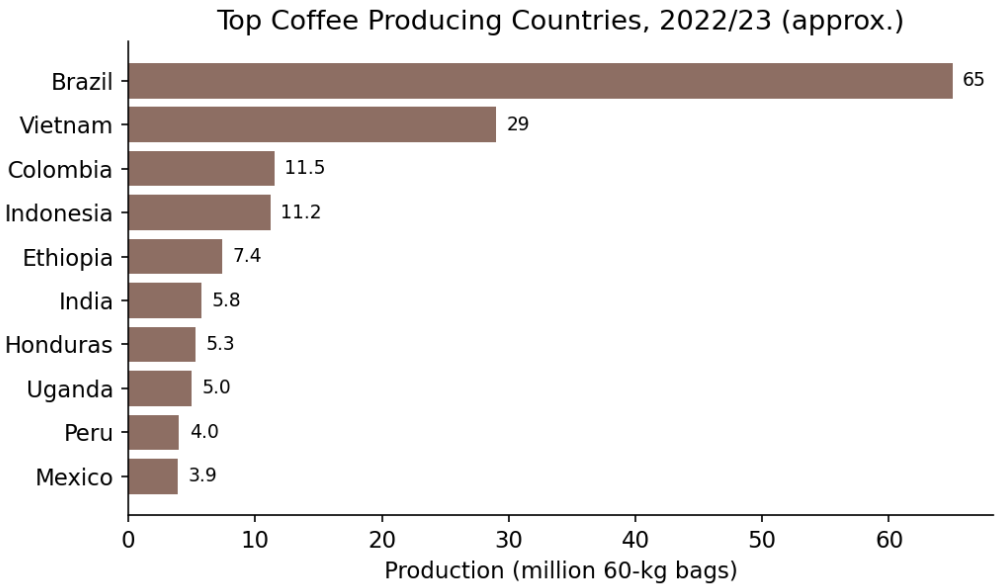


Figure 1.1: Top coffee producing countries in 2022/23, in million 60-kg bags (approximate).

This document is a structured reference on the global coffee industry. It covers botany, growing regions, processing, roasting, brewing, chemistry, trade and prices, sustainability, health, culture and market trends. It contains tables and figures so that it can be used to evaluate document-understanding and retrieval systems. All numerical values are rounded approximations for illustration.

1.1 Coffee in numbers

Table 1.1: Coffee at a glance (approximate, 2022/23)

Indicator	Value
World production	≈ 168 million 60-kg bags
World consumption	≈ 170 million 60-kg bags
Leading producer	Brazil (≈ 39% of world output)
Second-largest producer	Vietnam (≈ 17%)
Share of arabica in production	≈ 58%
Share of robusta in production	≈ 42%
Largest importing region	European Union
Largest single-country consumer	United States
Weight of one standard bag	60 kg
Number of coffee-growing countries	≈ 80

1.2 A brief timeline

Table 1.2: Key events in coffee history

Period	Event
c. 9th century	Legendary discovery of coffee in Ethiopia
15th century	Coffee drinking spreads in Yemen
1554	First coffeehouses open in Constantinople
1650	First coffeehouse in England (Oxford)
1720s	Coffee plants taken to the Caribbean and Brazil
1822	Brazil gains independence and expands coffee farming
1901	First patent for instant coffee (Satori Kato, Chicago)
1938	Nescafé launched
1971	First Starbucks store opens in Seattle
2000s	Specialty coffee and 'third wave' movement grows

2. Botany and Species

Coffee plants belong to the family Rubiaceae. More than 120 species of Coffea are known, but only two account for almost all commercial production: Coffea arabica (arabica) and Coffea canephora (robusta). A third species, Coffea liberica, is grown in small quantities in Southeast Asia, while Coffea eugenioides and others are used mainly in breeding.

Arabica is a tetraploid species with 44 chromosomes, a self-pollinating plant that originated from a natural cross between Coffea canephora and Coffea eugenioides. Robusta is diploid with 22 chromosomes and is cross-pollinating, which gives it greater genetic diversity. Arabica trees grow best at altitudes of 900–2,000 metres with temperatures of 18–22 °C, while robusta tolerates lower altitudes between 0 and 800 metres and temperatures of 22–30 °C.

Table 2.1: Comparison of arabica and robusta

Characteristic	Arabica	Robusta
Chromosomes	44 (tetraploid)	22 (diploid)
Ideal altitude	900–2,000 m	0–800 m
Ideal temperature	18–22 °C	22–30 °C
Annual rainfall	1,500–2,000 mm	2,000–3,000 mm
Caffeine content	1.0–1.5 %	2.0–2.7 %
Flavour	Sweet, acidic, complex	Bitter, earthy, strong
Disease resistance	Low (leaf rust)	High
Time to first harvest	3–4 years	2–3 years
Typical yield (kg/ha)	800–2,000	1,500–3,000
Price	Higher	Lower

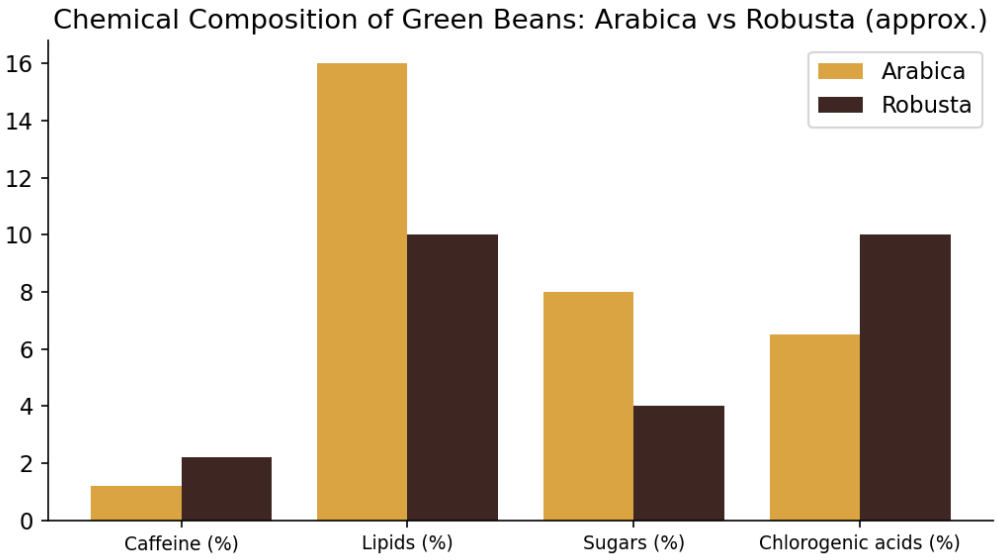


Figure 2.1: Approximate chemical composition of green arabica and robusta beans.

2.1 Varieties and cultivars

Within arabica there are many varieties, including Typica, Bourbon, Caturra, Catuai, Geisha (Gesha) and SL28. Typica and Bourbon are the historic parent lines. Geisha, originally from Ethiopia and made famous by Panama's Hacienda La Esmeralda, has fetched record auction prices above USD 1,000 per pound for its floral and tea-like profile. Hybrid varieties such as Catimor and Sarchimor combine arabica quality with leaf-rust resistance from the Timor Hybrid.

2.2 The coffee cherry

The fruit of the coffee plant is a cherry, usually containing two seeds with flat sides facing each other. From outside to inside, the layers are the skin (exocarp), the pulp (mesocarp), the mucilage, the parchment (endocarp), the silverskin and the bean itself. A single tree produces about 2–4 kg of cherries per year, which corresponds to around 0.5 kg of green coffee. About 4,000 hand-picked cherries are needed to make 1 kg of roasted coffee, which underlines how labour-intensive the harvest is.

3. Growing Regions and Production

Coffee is grown in about 80 countries across Latin America, Africa and Asia-Pacific. Each region contributes distinctive flavour profiles because of differences in altitude, soil, climate and processing traditions. Brazil alone produces nearly 40% of the world's coffee, mostly in the states of Minas Gerais, Espírito Santo and São Paulo, while Vietnam is the leading robusta producer, concentrated in the Central Highlands around Buôn Ma Thu■t.

Table 3.1: Production by country, 2022/23 (million 60-kg bags, approximate)

Country	Production	Main species	Typical flavour notes
Brazil	65.0	Arabica / robusta (conilon)	Nutty, chocolate, low acidity
Vietnam	29.0	Robusta	Strong, bitter, full body
Colombia	11.5	Arabica	Balanced, caramel, mild acidity
Indonesia	11.2	Robusta / arabica	Earthy, spicy, heavy body
Ethiopia	7.4	Arabica	Floral, fruity, tea-like
India	5.8	Robusta / arabica	Spicy, mellow
Honduras	5.3	Arabica	Sweet, chocolate
Uganda	5.0	Robusta / arabica	Earthy, fruity
Peru	4.0	Arabica	Mild, nutty
Mexico	3.9	Arabica	Light, clean

3.1 Latin America

Latin America produces about 60% of the world's coffee. Colombia is famous for its washed arabicas and for the Federación Nacional de Cafeteros, a cooperative organisation that represents more than 540,000 coffee-growing families and manages the 'Juan Valdez' brand. Central American countries such as Guatemala, Costa Rica and Honduras grow high-altitude coffees with bright acidity, while Costa Rica is known for banning robusta cultivation by law since 1989.

3.2 Africa

Africa is the birthplace of arabica. Ethiopia, with its thousands of local heirloom varieties, produces both natural-processed coffees from Yirgacheffe and Sidamo and washed coffees with jasmine and citrus notes. Kenya is known for SL28 and SL34

varieties with intense blackcurrant-like acidity, auctioned weekly in Nairobi. Uganda is Africa's largest robusta exporter. Coffee is Ethiopia's most important export, representing roughly one third of foreign-currency earnings.

3.3 Asia-Pacific

Vietnam became the world's second-largest producer after liberalisation in the 1990s, increasing output from about 1.5 million bags in 1990 to almost 30 million. Indonesia produces specialty coffees such as Sumatra Mandheling and the controversial Kopi Luwak. Yemen, the historic origin of the coffee trade, still grows small quantities of heritage coffees on terraced mountainsides. Papua New Guinea and India (Monsooned Malabar) offer additional distinctive profiles.

Table 3.2: Notable coffee origins and their characteristics

Origin	Region	Altitude (m)	Processing	Cup profile
Yirgacheffe	Ethiopia	1,700–2,200	Washed / natural	Floral, lemon, bergamot
Huila	Colombia	1,500–1,900	Washed	Caramel, red fruit
Antigua	Guatemala	1,500–1,700	Washed	Chocolate, spice
Kona	Hawaii, USA	150–900	Washed	Smooth, nutty
Blue Mountain	Jamaica	900–1,700	Washed	Mild, clean
Mandheling	Sumatra	750–1,500	Wet-hulled	Earthy, herbal
Kiambu	Kenya	1,400–2,000	Washed	Blackcurrant, tomato

4. From Cherry to Green Coffee: Harvesting and Processing

After flowering, coffee cherries ripen in six to nine months. Harvesting is done either by selective picking, in which only ripe red cherries are collected by hand in several passes, or by strip picking, in which all cherries are stripped from the branch at once. Mechanical harvesting is used on flat Brazilian plantations but is impossible on steep slopes. Selective picking gives higher quality but demands more labour.

Processing removes the layers surrounding the bean. The method strongly influences the flavour, and there are three main approaches, summarised in Figure 4.1: washed (wet), natural (dry) and honey (pulped natural).

Figure: Three Main Coffee Processing Methods

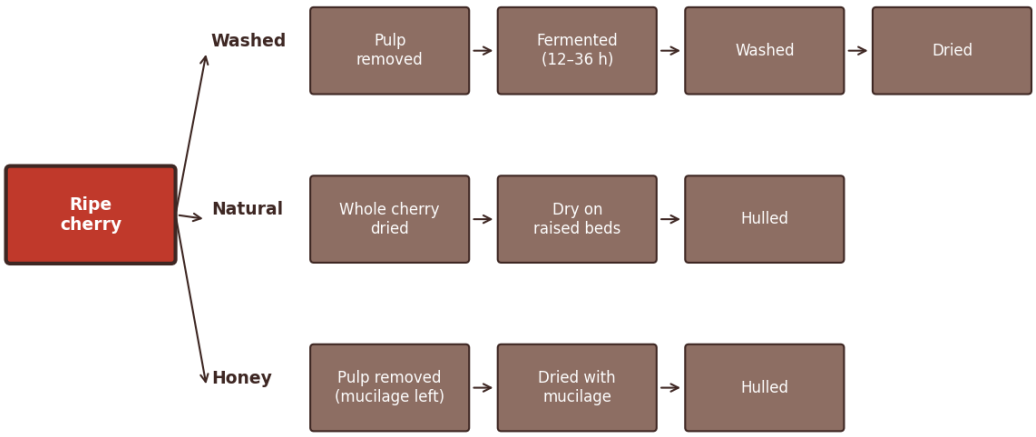


Figure 4.1: Three main coffee processing methods.

Table 4.1: Comparison of processing methods

Method	Water use	Duration	Flavour effect	Common in
Washed	High (up to 40 L per kg)	8–14 days	Clean, bright acidity	Colombia, Kenya, Central America
Natural	Low	3–5 weeks	Fruity, heavy body, sweet	Ethiopia, Brazil, Yemen
Honey	Medium	10–20 days	Sweet, balanced	Costa Rica, El Salvador
Wet-hulled (Giling Basah)	Medium	5–10 days	Earthy, low acidity	Indonesia
Anaerobic fermentation	Low–medium	2–7 days (tank)	Winey, intense, experimental	Specialty micro-lots

4.1 Drying, hulling and grading

After processing, coffee is dried to a moisture content of about 10–12% on raised African beds, concrete patios or in mechanical dryers. Drying too fast causes cracks, while slow drying risks mould. Dried coffee in parchment is stored for 4–8 weeks before hulling, which removes the parchment layer. Beans are then sorted by size (screen size 14 to 19, in 1/64-inch units), density and colour, and defects are removed by hand or by electronic sorters.

4.2 Green coffee storage and transport

Green coffee is shipped in 60-kg jute or sisal bags, often lined with GrainPro to protect against moisture, or in bulk liners inside 20-foot containers that hold about 275 bags (16.5 tonnes). Coffee is hygroscopic and absorbs odours, so storage conditions of 20 °C and 60% relative humidity are ideal. Properly stored green coffee keeps quality for about 12 months.

Table 4.2: Typical coffee grade classification (SCA green grading, 350 g sample)

Grade	Category 1 defects (full)	Category 2 defects	Description
Specialty (Grade 1)	0	≤ 5	Free of primary defects, ≥ 80 cupping points
Premium (Grade 2)	≤ 8	≤ 8	Fewer than 9 full defects, good cup
Exchange (Grade 3)	9–23	–	Standard commercial coffee
Below standard (Grade 4)	24–86	–	Low quality
Off grade (Grade 5)	> 86	–	Not exportable as coffee

5. Roasting

Roasting transforms green coffee into the aromatic brown beans that are brewed. During roasting, beans lose about 12–20% of their weight as water and gases, expand in volume by up to 80%, and develop over 800 volatile aroma compounds through the Maillard reaction, caramelisation and Strecker degradation. A typical roast lasts 9–15 minutes and reaches a final temperature of 190–240 °C.

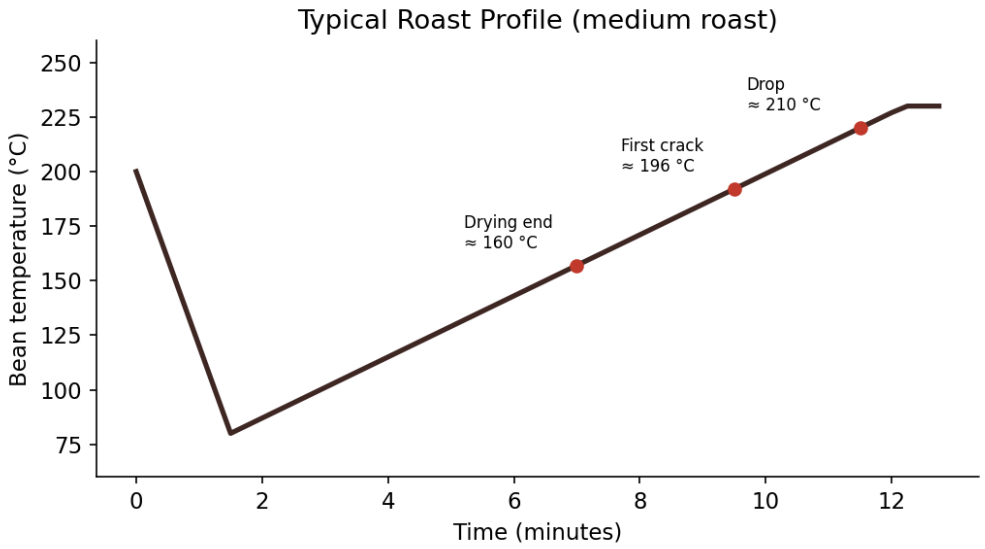


Figure 5.1: Typical roast profile showing bean temperature during a medium roast.

Two audible events mark the process. The first crack occurs at around 196 °C, when steam pressure fractures the bean with a popping sound, and marks the start of light roasts. The second crack occurs near 224 °C and is associated with oils appearing at the surface, typical for dark roasts. Roasters monitor the rate of rise (RoR), airflow and drum speed to achieve consistent results.

Table 5.1: Roast levels and their characteristics

Roast level	Drop temp. (°C)	Colour	Acidity	Body	Typical use
Light (Cinnamon)	195–205	Light brown	High	Light	Filter, specialty
Medium (City)	210–220	Medium brown	Medium	Medium	Drip, all-purpose
Medium-dark (Full City)	225–230	Dark brown	Low	Heavy	Espresso
Dark (French)	235–240	Very dark, oily	Very low	Heavy	Traditional espresso

Roast level	Drop temp. (°C)	Colour	Acidity	Body	Typical use
Italian	> 240	Almost black	Minimal	Thin, smoky	Moka pot, Italian style

5.1 Roasting equipment

Drum roasters heat beans in a rotating drum and are the most common type, from 1-kg sample roasters to 600-kg industrial machines. Fluid-bed (hot-air) roasters suspend beans in a stream of hot air and produce cleaner, brighter cups. Afterburners reduce smoke emissions and energy recovery systems cut gas consumption. After roasting, beans must be cooled rapidly within 4 minutes to stop the reaction.

5.2 Freshness and degassing

Roasted coffee releases carbon dioxide for days after roasting. Espresso is usually best 5–14 days after roasting, whereas filter coffee needs 3–10 days. Staling is accelerated by oxygen, moisture, heat and light, and ground coffee loses most of its aroma within 30 minutes. One-way valve bags let CO2 escape without allowing oxygen in. Whole beans stay at good quality for about 4–6 weeks after roasting when stored in an airtight container.

6. Brewing and Preparation

Brewing is the extraction of soluble compounds from ground coffee with hot water. A typical extraction removes 18–22% of the coffee mass, which is regarded as the ideal range. Under-extraction (below 18%) tastes sour and weak, while over-extraction (above 22%) tastes bitter and harsh. The key variables are grind size, water temperature, contact time, brew ratio and agitation.

Table 6.1: Brewing methods and recommended parameters

Method	Grind	Water temp. (°C)	Brew ratio (coffee:water)	Time
Espresso	Fine	90–96	1:2 (18 g in, 36 g out)	25–30 s
Pour-over (V60)	Medium-fine	92–96	1:15 to 1:16	2:30–3:30 min
French press	Coarse	93–96	1:15	4 min
AeroPress	Medium-fine	80–90	1:12 to 1:16	1–2 min
Moka pot	Fine-medium	Near boiling	1:7 to 1:10	4–5 min
Cold brew	Coarse	Room / fridge	1:5 to 1:8 (concentrate)	12–18 h
Turkish / Arabic (ibrik)	Extra fine	Near boiling	1:10	3–4 min

6.1 Espresso

Espresso is made by forcing hot water at about 9 bar of pressure through finely ground coffee. The first modern espresso machine was patented by Angelo Moriondo in Turin in 1884, and the lever machine invented by Achille Gaggia in 1948 introduced crema. A standard double shot uses 18 g of ground coffee to produce roughly 36 g of liquid in 25–30 seconds. Espresso is the base for most café drinks.

Table 6.2: Common espresso-based drinks

Drink	Espresso	Milk	Foam	Typical size
Espresso	1 shot	–	–	30 ml
Americano	1–2 shots	–	–	240 ml (hot water)
Cappuccino	1 shot	Steamed, 1/3	1/3 foam	150–180 ml
Flat white	2 shots (ristretto)	Microfoam	Thin	160–180 ml
Latte	1 shot	Steamed, 2/3	Thin layer	240–350 ml

Drink	Espresso	Milk	Foam	Typical size
Macchiato	1 shot	–	Dollop of foam	60 ml
Mocha	1 shot	Steamed + chocolate	Optional cream	240–350 ml

6.2 Traditional methods in the Middle East

In Egypt and the Levant, coffee is traditionally prepared in a small long-handled pot called a kanaka or ibrik. Finely ground coffee, often with cardamom, is boiled slowly with water and sugar and served unfiltered in small cups. The sweetness level is specified when ordering: sada (no sugar), mazbout (medium) or ziyada (extra sugar). Egyptian coffeehouses (ahwa) have been social centres for centuries, where people play backgammon (tawla) and discuss daily life.

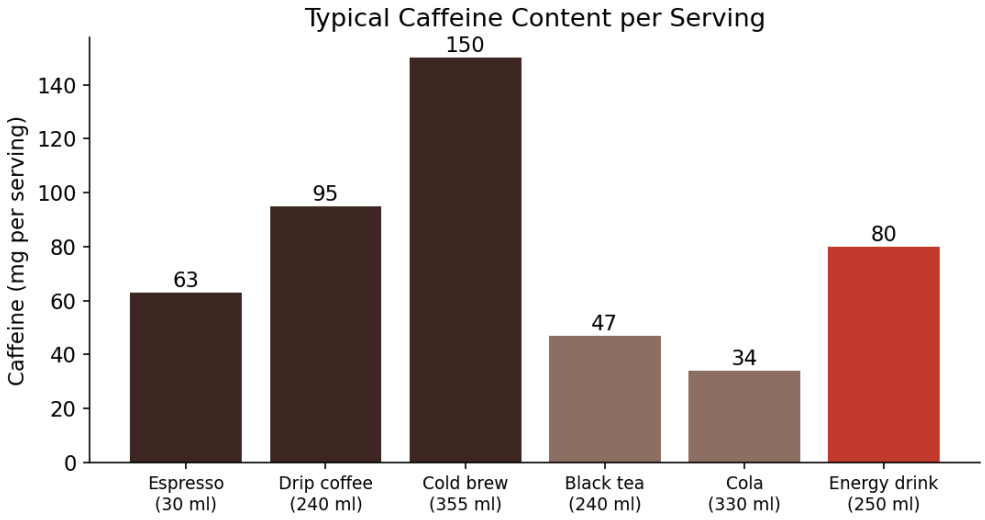


Figure 6.1: Typical caffeine content of common beverages per serving.

7. The Chemistry of Coffee and Health

Roasted coffee contains more than 1,000 identified chemical compounds. Major components include carbohydrates, lipids, proteins, chlorogenic acids, trigonelline, melanoidins and caffeine. Caffeine (1,3,7-trimethylxanthine) is a central nervous system stimulant that blocks adenosine receptors in the brain, reducing the sensation of tiredness. Its half-life in healthy adults is about 5 hours, with considerable individual variation.

Table 7.1: Approximate composition of roasted arabica beans (% of dry weight)

Component	Content (%)	Role
Carbohydrates (polysaccharides)	≈ 38	Body, structure
Melanoidins	≈ 25	Colour, bitterness
Lipids	≈ 17	Mouthfeel, aroma carrier
Proteins and amino acids	≈ 7.5	Aroma precursors
Chlorogenic acids	≈ 3	Acidity, antioxidants
Minerals	≈ 4.5	Potassium, magnesium
Trigonelline	≈ 1	Converted to niacin
Caffeine	≈ 1.2	Stimulant, bitterness

7.1 Health effects

Large observational studies suggest that moderate coffee consumption of three to four cups per day is associated with a lower risk of type 2 diabetes, Parkinson's disease, liver cirrhosis and some cancers such as liver and endometrial cancer, and with no increase in cardiovascular risk for most adults. These are associations, not proof of causation. Unfiltered coffee contains diterpenes (cafestol and kahweol) that can raise LDL cholesterol.

The European Food Safety Authority considers that up to 400 mg of caffeine per day (about four cups of filter coffee) is safe for healthy adults, and up to 200 mg per day for pregnant women. Excessive intake can cause anxiety, insomnia, palpitations and digestive upset. People with specific medical conditions should follow the advice of their physician.

Table 7.2: Caffeine guidelines

Group	Suggested daily limit	Source
Healthy adults	≤ 400 mg	EFSA, FDA
Pregnant / breastfeeding women	≤ 200 mg	EFSA, WHO advice
Children and adolescents	≤ 3 mg per kg body weight	EFSA
Single dose	≤ 200 mg	EFSA

7.2 Decaffeination

Decaffeinated coffee has at least 97% of its caffeine removed. Common methods include the Swiss Water Process, which uses water and activated carbon; supercritical CO2 extraction; and solvent-based methods using ethyl acetate or methylene chloride. The process is applied to green beans before roasting. Decaf contains about 2–15 mg of caffeine per cup.

8. Global Trade, Prices and Economics

Coffee is traded on two main futures exchanges. Arabica futures, known as the 'C' contract, are traded on the Intercontinental Exchange (ICE) in New York in units of 37,500 pounds. Robusta is traded on ICE Europe in London in lots of 10 tonnes. Prices are quoted in US cents per pound for arabica and US dollars per tonne for robusta. Physical prices equal the futures price plus or minus a differential that reflects quality and origin.

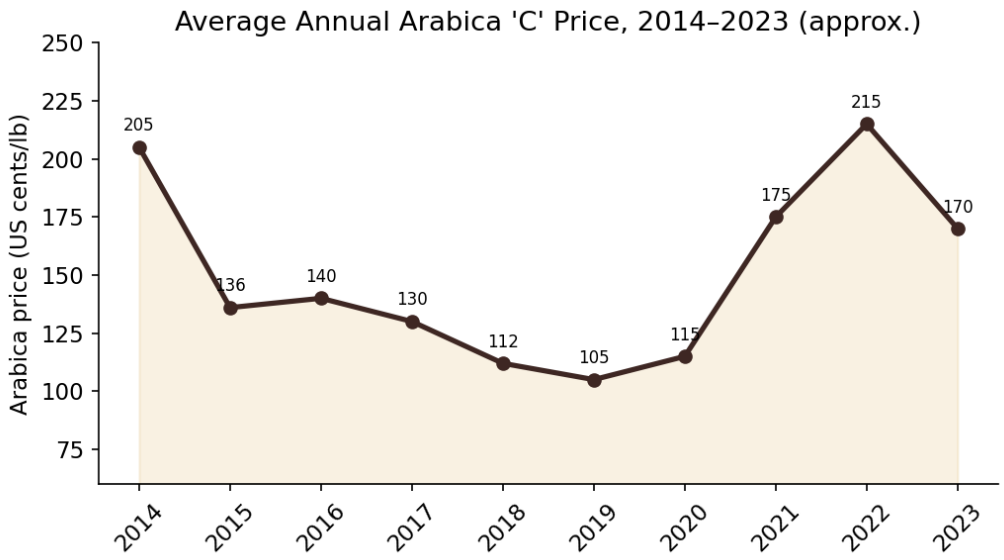


Figure 7.1: Average annual arabica 'C' price, 2014–2023 (approximate).

Prices are volatile because supply depends on weather, particularly frost and drought in Brazil. The July 2021 frost in Minas Gerais, the worst in decades, triggered a price surge. Speculative funds, the Brazilian real exchange rate and stock levels in certified warehouses also influence the market.

Table 8.1: Largest coffee importers (approximate, million 60-kg bags)

Importer	Imports	Main sources
European Union (bloc)	≈ 45	Brazil, Vietnam, Uganda
United States	≈ 25	Brazil, Colombia, Vietnam
Japan	≈ 7	Brazil, Vietnam, Colombia
United Kingdom	≈ 5	Brazil, Vietnam, Honduras
Canada	≈ 4.5	Colombia, Brazil, Vietnam
Russia	≈ 4	Vietnam, Brazil, Uganda

8.1 Who earns what?

Smallholders produce roughly 80% of the world's coffee on farms of less than five hectares. Farmers typically receive only a small share of the retail price, as shown in Figure 8.1. Value is added mainly in roasting, branding and retail. Direct trade, cooperatives and Fairtrade minimum prices aim to improve farmers' income. The Fairtrade minimum price for washed arabica was USD 1.80 per pound including a premium of 20 cents in 2023.

Figure: The Coffee Value Chain

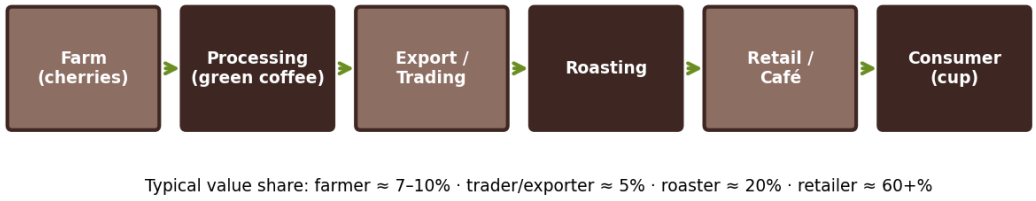


Figure 8.1: The coffee value chain and typical value shares.

Table 8.2: Illustrative cost of production for a smallholder farm (per hectare per year)

Cost item	Amount (USD)	Share (%)
Labour (harvest and maintenance)	1,100	55
Fertiliser and chemicals	420	21
Processing and transport	260	13
Equipment and depreciation	140	7
Other costs	80	4
Total	2,000	100

9. Sustainability and Climate Change

Coffee is highly sensitive to climate. Arabica needs a narrow temperature range, and rising temperatures, irregular rainfall and increased pests such as coffee leaf rust (Hemileia vastatrix) and the coffee berry borer threaten production. Studies project that the area suitable for arabica could fall by 50% or more by 2050 under high-emission scenarios, as illustrated in Figure 9.1.

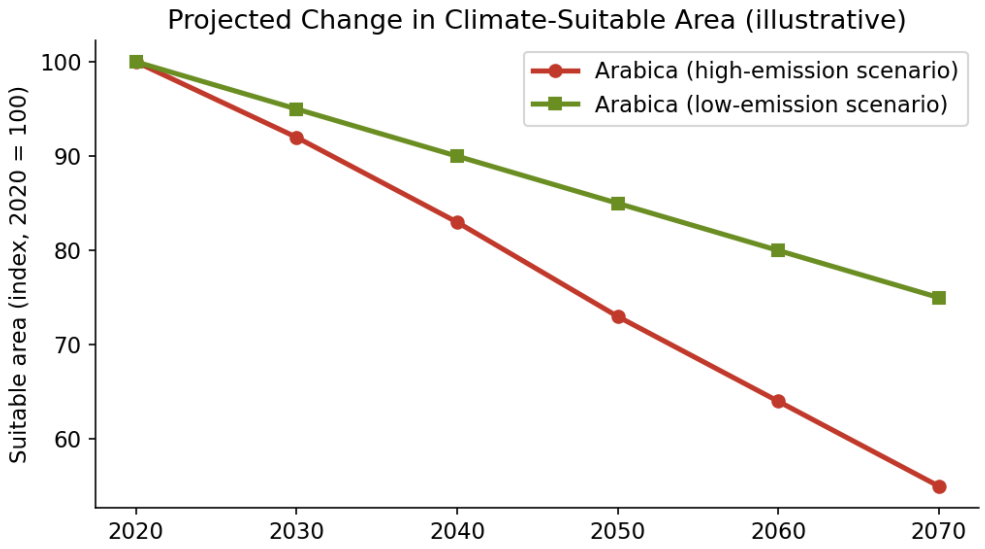


Figure 9.1: Projected change in climate-suitable area for arabica (illustrative scenarios).

Coffee leaf rust devastated Ceylon in the 1870s, turning it into a tea-producing country. A major outbreak in Central America in 2012–2013 reduced production by up to 30% in some countries and displaced thousands of workers. Adaptation strategies include shade-grown agroforestry, resistant varieties, moving to higher altitudes and improved water management.

Table 9.1: Sustainability certification programmes

Programme	Focus	Key feature
Fairtrade	Farmer income and community	Minimum price and social premium
Rainforest Alliance	Environment and livelihoods	Includes UTZ since 2018 merger
Organic	No synthetic chemicals	Certified by national regulators
4C / Coffee Sustainability Code	Baseline sustainability	Entry-level standard
Bird Friendly	Biodiversity	Shade-grown criteria (Smithsonian)

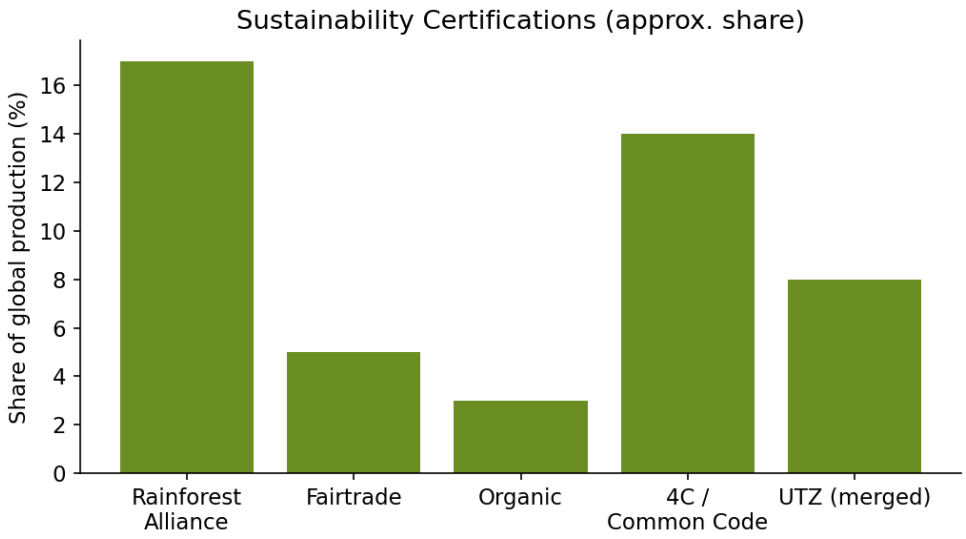


Figure 9.2: Approximate share of global production under major certification schemes.

9.1 Environmental footprint

The carbon footprint of coffee is dominated by farming (fertiliser use and land-use change) and by milk in café drinks. A cup of black coffee has a footprint of roughly 50–100 grams of CO2e, while a latte can emit 340 grams or more. The European Union Deforestation Regulation (EUDR) requires importers to prove that coffee is not linked to deforestation after 2020. Wet processing also generates wastewater that must be treated, and spent coffee grounds can be composted or used for biofuels.

9.2 Social issues

Many smallholders live below the poverty line, face price volatility and have limited access to credit. Child labour and labour exploitation have been documented in some regions. Ageing farmer populations and young people leaving rural areas threaten the future supply of coffee, and the sector is therefore investing in income diversification and youth programmes.

10. Coffee Culture and Cafés

Coffee has shaped social life for centuries. In 17th-century England coffeehouses were called 'penny universities' because for the price of a cup of coffee one could join discussions among scholars and merchants. Lloyd's of London began as a coffeehouse where shipowners and insurers met. In Italy, espresso bars developed a culture of quick standing coffee, while in Scandinavia the tradition of fika is a daily social break.

10.1 The three waves of coffee

Table 10.1: The three waves of coffee

Wave	Period	Characteristics	Examples
First wave	1800s–1960s	Mass consumption, convenience, vacuum-packed ground coffee, instant	Folgers, Maxwell House, Nescafé
Second wave	1960s–2000s	Espresso drinks, café culture, origin awareness	Starbucks, Peet's, Costa
Third wave	2000s–today	Specialty, direct trade, light roasts, brewing precision	Stumptown, Blue Bottle, Tim Wendelboe

10.2 The modern café business

Cafés are among the most common small businesses. A typical independent café in a city earns about 60–70% of revenue from beverages, with a gross margin of 70–80% on coffee drinks after ingredient costs, but high rent and labour reduce net profit to 5–15%. Chains dominate the market: Starbucks has more than 38,000 stores worldwide, Luckin Coffee in China passed 20,000 stores in 2023, and Costa, Dunkin' and Tim Hortons also have large networks.

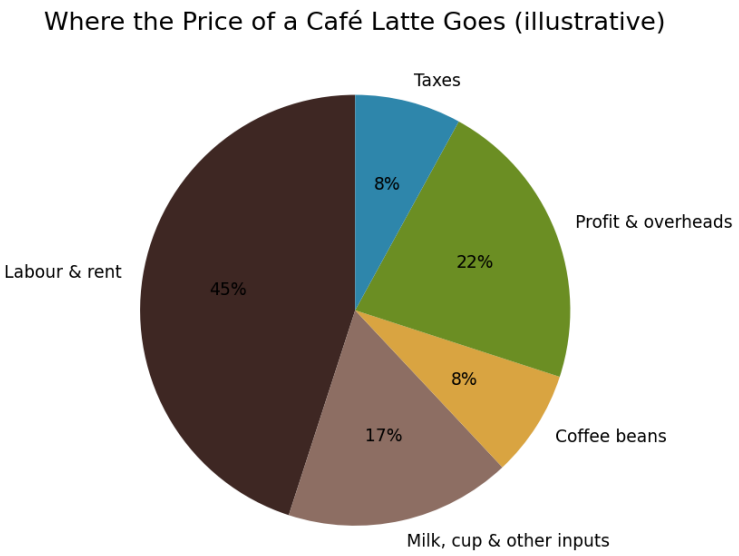


Figure 10.1: Where the price of a café latte goes (illustrative).

10.3 Coffee in Egypt

Egypt has a long coffee-drinking tradition dating back to the 16th century, when Yemeni merchants introduced coffee to Cairo and the Al-Azhar quarter. Cairo's historic coffeehouse El Fishawy in Khan el-Khalili has served customers since 1773. Egypt imports most of its coffee, mainly from Brazil, Ethiopia and Vietnam. Tea remains more consumed per capita, but coffee shop chains and specialty cafés have grown quickly in Cairo, Alexandria and new urban communities, driven by a young population.

Table 10.2: Famous historic coffeehouses

Coffeehouse	City	Founded	Notes
Café Florian	Venice, Italy	1720	Oldest continuously operating café in Europe
Café Procope	Paris, France	1686	Meeting place of Enlightenment writers
El Fishawy	Cairo, Egypt	1773	Open almost 24 hours in Khan el-Khalili
Café Central	Vienna, Austria	1876	Intellectual café
Cafe Tortoni	Buenos Aires, Argentina	1858	Tango and arts

11. Market Trends and the Future of Coffee

Global coffee consumption has grown by about 2% per year over the last decade. Growth is strongest in Asia and in producing countries, whereas Europe and North America are mature markets. Key trends include premiumisation, ready-to-drink (RTD) cold coffee, single-serve capsules, sustainability transparency and digital ordering through mobile apps.

Table 11.1: Market segments and trends

Segment	Trend	Estimated growth (CAGR)
Specialty coffee	Rising consumer demand for origin and quality	≈ 8–10 %
Ready-to-drink	Cold brew, nitro, canned coffee	≈ 8 %
Single-serve capsules	Convenience, recyclable capsules	≈ 5 %
Instant coffee	Premium freeze-dried, growth in Asia	≈ 3 %
Decaf	Health-focused consumers	≈ 4 %
Plant-based milk drinks	Oat, soy and almond lattes	≈ 10 %

11.1 Technology and innovation

Technology is changing the coffee sector in multiple ways: precision fermentation, AI-assisted roast profiling, blockchain-based traceability, smart espresso machines with pressure profiling, and robotic baristas. Lab-grown coffee and 'beanless' coffee made from upcycled seeds are emerging as alternatives, motivated by supply and climate risks. New varieties such as Coffea stenophylla, a heat-tolerant wild species rediscovered in Sierra Leone in 2018, may help adapt to warming.

11.2 Challenges ahead

- Climate change shrinking suitable growing areas for arabica
- Price volatility and farmer poverty
- Ageing farmers and lack of generational renewal
- Regulatory pressure on deforestation and traceability
- Rising costs of labour, fertiliser and shipping

11.3 Conclusions

Coffee is more than a beverage: it is a global industry connecting millions of farmers to billions of consumers. Its future depends on making production climate-resilient and fair, while keeping the quality that consumers value. Producers, roasters, retailers, governments and consumers all share responsibility for a sustainable supply chain.

Table 11.2: Glossary

Term	Definition
Arabica	Coffea arabica, the higher-quality species, about 58% of production
Robusta	Coffea canephora, hardy and high in caffeine
Crema	Golden foam on top of an espresso
Cupping	Standardised sensory evaluation of coffee
SCA score	Specialty Coffee Association cupping score; 80+ is specialty
Differential	Premium or discount versus futures price
Bag	Standard unit of 60 kg of green coffee
RoR	Rate of rise: temperature increase per minute in roasting
Agroforestry	Growing coffee under shade trees
Fika	Swedish tradition of a coffee break with pastries

12. Quality Evaluation and Cupping

Coffee quality is evaluated through a mix of physical analysis of green beans and sensory evaluation of roasted samples. The standard sensory method is called cupping. Roasted coffee is ground, placed in identical cups, covered with hot water at about 93 °C and left to steep for four minutes before the crust is broken and the coffee is evaluated by smelling and slurping with a spoon. Professional tasters, known as Q Graders, are certified by the Coffee Quality Institute (CQI) after passing a series of 22 tests.

The Specialty Coffee Association (SCA) protocol scores ten attributes on a scale up to 10 points each, for a total of 100 points. Coffees scoring 80 or above are considered specialty grade. Scores between 85 and 89.99 are described as excellent and those from 90 upwards as outstanding. Figure 12.1 compares the attribute profiles of a specialty lot and a commercial lot.

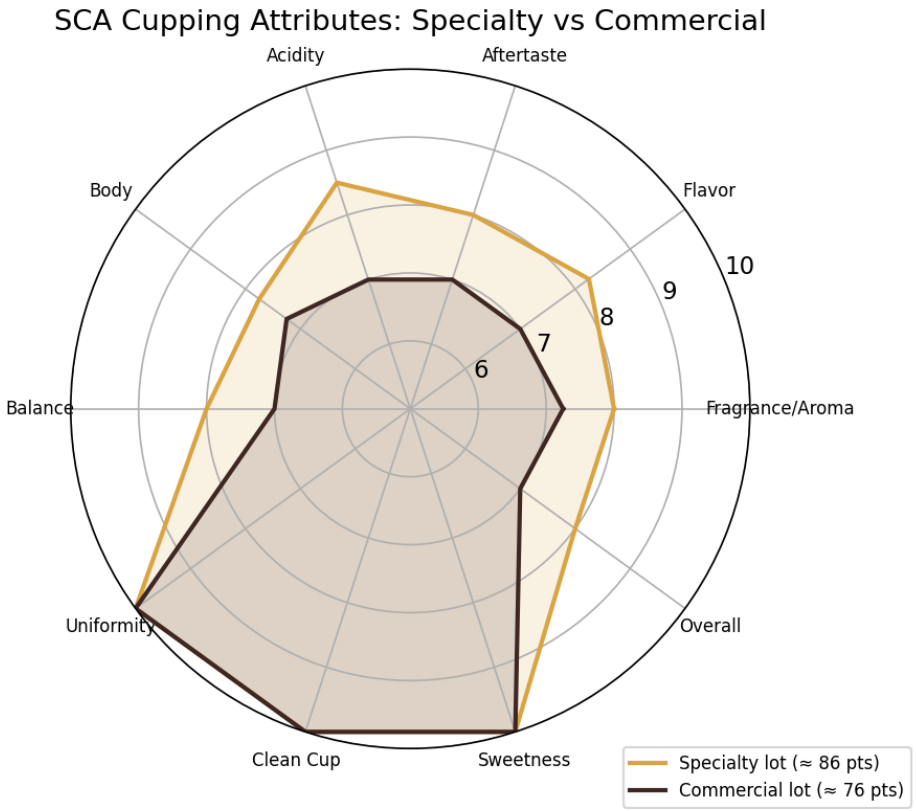


Figure 12.1: SCA cupping attribute scores for a specialty lot and a commercial lot (illustrative).

Table 12.1: SCA cupping score interpretation

Total score	Classification	Description
90–100	Outstanding	Exceptional specialty coffee
85–89.99	Excellent	Very high-quality specialty
80–84.99	Very good	Specialty threshold
Below 80	Not specialty	Commercial or premium grade

12.1 The SCA flavour wheel

The Coffee Taster's Flavor Wheel, developed by the SCA and updated with World Coffee Research in 2016, provides a common vocabulary for describing aroma and taste. It organises descriptors in nine main categories: fruity, sour/fermented, green/vegetative, other, roasted, spices, nutty/cocoa, sweet and floral. Each category is divided into more specific terms, such as blueberry, lemon, jasmine, almond, brown sugar and dark chocolate.

Table 12.2: Major categories of the flavour wheel with examples

Category	Example descriptors	Typical origin or process
Fruity	Blueberry, lemon, apple, cherry	Ethiopia natural, Kenya washed
Floral	Jasmine, rose, chamomile	Geisha, Yirgacheffe
Sweet	Honey, caramel, maple syrup	Honey process, Colombia
Nutty / cocoa	Almond, hazelnut, dark chocolate	Brazil natural
Spices	Cinnamon, clove, pepper	Indonesia, India
Roasted	Toast, tobacco, burnt sugar	Dark roasts
Sour / fermented	Winey, vinegar, overripe	Anaerobic or defective lots
Green / vegetative	Grassy, herbal, olive	Under-developed roasts

12.2 Physical analysis of green coffee

Buyers inspect a 350-gram sample of green beans for defects such as black beans, sour beans, insect damage, broken beans and foreign matter. Moisture content is measured with a meter and should be 10–12%. Water activity should stay below 0.65 to prevent mould growth. Bean density and screen size are also reported; a larger screen size such as 18 (7.1 mm) is generally associated with higher prices, although size alone does not determine cup quality.

Table 12.3: Typical green coffee specifications for export

Parameter	Specialty target	Commercial target
Moisture content	10–12 %	10–13 %
Water activity	0.50–0.60	< 0.70
Screen size	15 and above	14 and above
Defects per 350 g	≤ 5 secondary, 0 primary	≤ 23 full defects
Cupping score	≥ 80	70–79
Ochratoxin A limit (EU)	< 5 µg/kg	< 5 µg/kg

13. Coffee Traditions Around the World

Coffee preparation and etiquette differ greatly between cultures. Understanding these traditions helps explain why consumption patterns, serving sizes and flavour preferences vary across regions. In the Nordic countries, filter coffee is consumed in large quantities, whereas in southern Europe people prefer small, strong espresso. In East Asia, canned and iced coffee dominate convenience-store sales, and in Africa and the Middle East coffee is often a ritual of hospitality.

Table 13.1: Coffee traditions in selected countries

Country	Tradition	Typical serving
Ethiopia	Coffee ceremony (buna) with roasting, grinding and brewing in a jebena clay pot, three rounds: abol, tona, baraka	Small cups, three rounds
Italy	Espresso at the bar, cappuccino only before late morning	30 ml, standing
Turkey	Turkish coffee recognised by UNESCO in 2013, served with lokum	60–70 ml, unfiltered
Saudi Arabia / Gulf	Qahwa Arabiyya with cardamom and saffron, offered to guests	Tiny cups, poured by host
Vietnam	Cà phê s■a ■á: robusta dripped through a phin over condensed milk	150–200 ml iced
Japan	Kissaten culture, siphon and hand-drip, canned coffee vending machines	180–250 ml
Sweden	Fika: coffee break with cinnamon bun (kanelbulle)	Filter coffee, refills
Australia / NZ	Flat white and long black, strong café scene	160–180 ml
Egypt	Ahwa culture, mazbout coffee, Turkish-style coffee with cardamom	Small cups

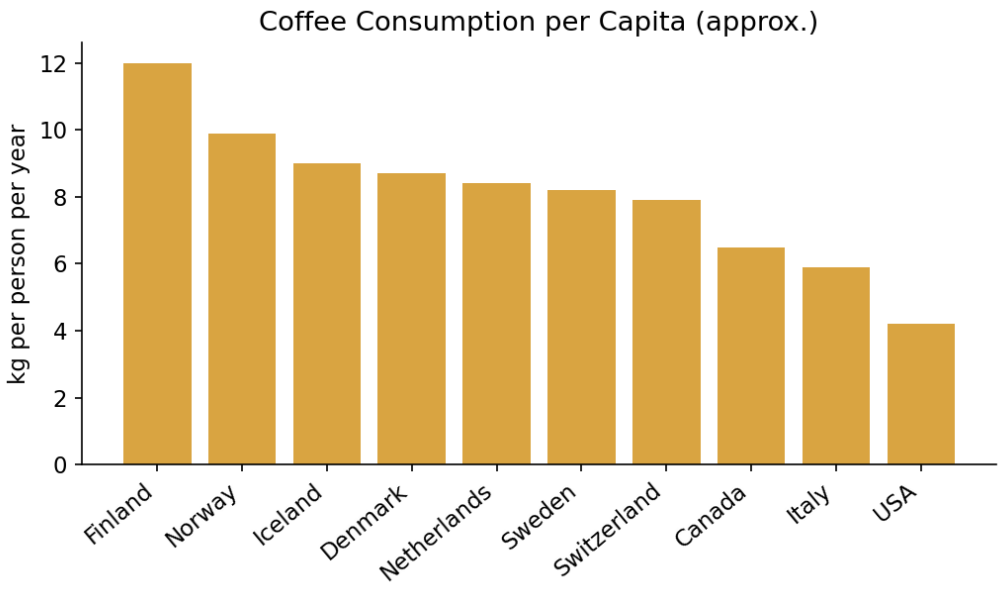


Figure 13.1: Coffee consumption per capita in selected countries (kg per person per year, approximate).

13.1 Coffee ceremonies

The Ethiopian coffee ceremony is a central social ritual. Green beans are washed and roasted on a pan over charcoal, and the aromatic smoke is wafted towards the guests. The beans are ground in a mortar (mukecha) and brewed in a round-bottomed clay pot called a jebena, with incense burning nearby. The ceremony can last an hour or more, and guests usually drink at least three cups. Similar hospitality rituals exist in Saudi Arabia, the UAE and Yemen, where serving coffee is a sign of respect, and guests signal that they have had enough by gently shaking the cup.

13.2 Emerging coffee markets

Coffee is growing fastest in markets with a young, urban population and rising incomes, such as China, India, Indonesia, Saudi Arabia, Nigeria and Egypt. Chinese per-capita consumption is still only about 10 cups per year, compared with more than 300 in many Western countries, which suggests large growth potential. Figure 13.2 illustrates the kind of steady import growth seen in such emerging markets.

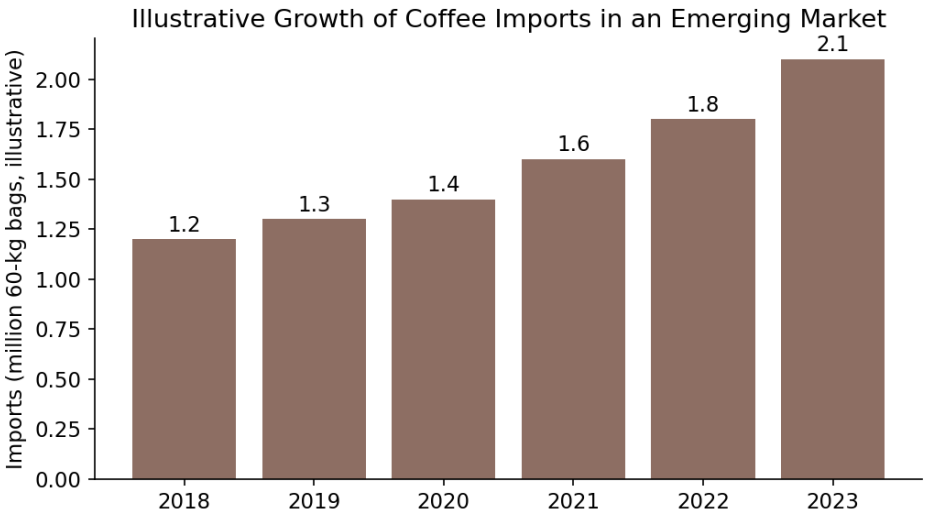


Figure 13.2: Illustrative growth of coffee imports in an emerging market.

Table 13.2: Estimated cups per person per year (approximate)

Country	Cups per person per year
Finland	≈ 1,000
Norway	≈ 830
Netherlands	≈ 700
Italy	≈ 500
United States	≈ 350
Japan	≈ 330
Brazil	≈ 500
China	≈ 10

14. Appendix: Worked Examples and Frequently Asked Questions

This appendix collects short worked examples and frequently asked questions. Each answer can be located in a single table or paragraph, which makes the section useful for testing question-answering over structured content.

A.1 Worked example: converting green coffee to roasted coffee

A roaster buys one 60-kg bag of green coffee. Roasting causes a weight loss of about 15%, so the output is 60 × 0.85 = 51 kg of roasted coffee. If each espresso uses 18 g of roasted beans, one bag yields about 2,833 espressos. At a green coffee cost of USD 6.00 per kg (USD 360 per bag), the bean cost per espresso is about USD 0.13 after roast loss.

Table A.1: Roasting yield and cost example

Item	Value
Green coffee purchased	60 kg
Green price	USD 6.00/kg
Total green cost	USD 360
Roast weight loss	15 %
Roasted coffee output	51 kg
Cost per kg roasted (beans only)	USD 7.06
Coffee per espresso	18 g
Number of espressos	≈ 2,833
Bean cost per espresso	≈ USD 0.127

A.2 Worked example: brew ratio and extraction

A barista prepares a pour-over using a 1:16 ratio. For a 320 ml cup, the amount of coffee needed is 320 / 16 = 20 g. If the resulting brew has a total dissolved solids (TDS) concentration of 1.35% and the brew weighs 300 g, the extraction yield is (300 × 1.35%) / 20 = 20.25%, which is inside the ideal 18–22% range.

Table A.2: Brew calculation inputs and results

Parameter	Value
Target brew volume	320 ml

Parameter	Value
Brew ratio	1:16
Coffee dose	20 g
Brew weight	300 g
TDS	1.35 %
Extraction yield	20.25 %
Result	Within 18–22 % target

A.3 Frequently asked questions

Table A.3: Frequently asked questions

Question	Answer
Which country produces the most coffee?	Brazil, with about 39% of world production.
What is the main difference between arabica and robusta?	Arabica is sweeter and more acidic with about 1.2% caffeine; robusta is stronger and more bitter with about 2.2% caffeine.
How much weight does coffee lose during roasting?	About 12–20%, typically 15%.
What is the ideal extraction yield?	18–22% of the coffee mass.
How many cherries make 1 kg of roasted coffee?	About 4,000 hand-picked cherries.
What is the safe daily caffeine limit for healthy adults?	Up to 400 mg per day.
Where does the first crack happen?	At about 196 °C bean temperature.
What pressure is used for espresso?	About 9 bar.
What is the Coffee Belt?	The tropical zone between the Tropic of Cancer and Tropic of Capricorn where coffee grows.
What is a 'bag' of coffee?	A standard unit of 60 kg of green coffee.

A.4 Summary of key numbers

Table A.4: Quick-reference numbers

Quantity	Value
World production (2022/23)	≈ 168 million bags
Brazil share of production	≈ 39 %
Arabica share	≈ 58 %
Cups consumed per day	≈ 2.2 billion

Quantity	Value
Espresso pressure	≈ 9 bar
Ideal extraction	18–22 %
Roast weight loss	≈ 15 %
Daily caffeine limit (adults)	400 mg
Smallholders' share of production	≈ 80 %
Starbucks stores worldwide	> 38,000